

A logarithmic improvement in the lower bound for every die

Research note: endpoint odds, finite moments, and Laguerre tests

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Status. Complete proof, with three independent line-by-line mathematical reviews completed. The endpoint-odds representation and the bad-prefix estimate were developed in the parallel note `../independent_directions/endpoint_poisson_rigidity.tex`. The finite-moment recurrence and the tapered Laguerre test below were independently derived by collaborating agents. No formal verification is claimed.

Theorem 1. *There are absolute constants $c > 0$ and m_0 with the following property. Let $p_1, \dots, p_m$ be nondecreasing functions from $\{1, \dots, K\}$ to $[0, 1]$, and let $w_q > 0$ be probability weights. Suppose*

$$\mathbb{E}_w \prod_{j \in S} p_j = \frac{1}{|S| + 1} \quad (S \subseteq [m]). \quad (1)$$

If $m \geq m_0$, then the last and first row weights satisfy

$$w_K \leq \frac{1}{cm \log m}, \quad w_1 \leq \frac{1}{cm \log m}.$$

In particular, an equal-weight representation has

$$K \geq cm \log m.$$

Corollary 2. *Every die in every permutation-fair collection of n finite equiprobable dice has at least $cn \log n$ faces, for an absolute $c > 0$ and all sufficiently large n . Thus the exceptional-die optimum satisfies*

$$cn \log n \leq g(n) \leq Cn^2,$$

where the upper bound is the independently established disjoint-bump construction. Neither bound settles the minimum total size.

Proof. For a distinguished die, list its K faces in increasing order and put $p_j(q) = \Pr(X_j < X_0 \mid X_0 \text{ is face } q)$ for each other die j . These functions are nondecreasing. Conditional independence and fairness of every restriction give (1), with $m = n - 1$ and $w_q = 1/K$. $\square$

Endpoint odds and quantitative moments

Put $A(q) = \prod_j p_j(q)$ and

$$\eta(q) = (m + 1)w_q A(q).$$

This is a probability measure by (1). On rows with $A(q) > 0$ write $r_j = (1 - p_j)/p_j$ and $\lambda = \sum_j r_j$. Give a zero entry infinite odds when defining bad rows; its η -mass is zero.

Lemma 3. *Set $\varepsilon = m^{-1/2}$, and let G consist of rows on which every $r_j < \varepsilon$. For each integer $0 \leq k \leq \sqrt{m}$, with $k < m$,*

$$M_k := \sum_{q \in G} \eta(q) \frac{\lambda(q)^k}{k!} \quad \text{satisfies} \quad |M_k - 1| \leq \frac{2k + 1}{\sqrt{m}}.$$

The measure restricted to G need not have total mass exactly one.

Proof. Define $B_q(z) = \prod_j (p_j(q) + z(1 - p_j(q)))$. Expanding (1) gives the coefficient identities

$$(m+1)\mathbb{E}_w B_q(z) = \sum_{k=0}^m z^k, \quad (2)$$

$$(m+1)\mathbb{E}_w \left[(1-p_j) \prod_{i \neq j} (p_i + z(1-p_i)) \right] = \frac{1}{m} \sum_{k=0}^{m-1} (k+1)z^k. \quad (3)$$

For example, the probability of a particular failure set of size s is $\int_0^1 t^{m-s}(1-t)^s dt = 1/((m+1)\binom{m}{s})$; summing over failure sets proves these identities.

The bad rows $D = G^c$ form an initial segment. If D is nonempty, choose a column j_* with odds at least ε at its last row. Monotonicity gives $p_{j_*} \leq (1 - p_{j_*})/\varepsilon$ throughout D . Consequently, as an inequality between nonnegative polynomial coefficients,

$$(m+1)\mathbb{E}_w[B_q(z)\mathbf{1}_D] \preceq \frac{\varepsilon^{-1} + z}{m} \sum_{k=0}^{m-1} (k+1)z^k.$$

The same assertion is immediate when D is empty. It includes rows with zero entries without dividing by them. Writing e_k for the elementary symmetric polynomial in the odds, (2) now implies, for $0 \leq k < m$,

$$1 - \frac{(k+1)/\varepsilon + k}{m} \leq a_k := \sum_{q \in G} \eta(q) e_k(r(q)) \leq 1. \quad (4)$$

On a good row, expand λ^k as a sum over ordered k -tuples of indices. The terms with distinct indices sum to $k!e_k$. A union bound over a repeated pair of positions gives, for $k \geq 2$,

$$0 \leq \lambda^k - k!e_k \leq \binom{k}{2} \left(\sum_j r_j^2 \right) \lambda^{k-2} \leq \binom{k}{2} \varepsilon \lambda^{k-1}.$$

Thus $0 \leq M_k - a_k \leq ((k-1)\varepsilon/2)M_{k-1}$; for $k=1$ there is equality $M_1 = a_1$. Starting with $M_0 \leq 1$, induction gives $M_k \leq 1 + k\varepsilon$ whenever $k\varepsilon \leq 1$: the preceding moment is at most two, so $M_k \leq 1 + (k-1)\varepsilon$. The lower bound in (4) and $M_k \geq a_k$ give

$$M_k \geq 1 - (k+1)m^{-1/2} - k/m \geq 1 - (2k+1)m^{-1/2}.$$

This proves the lemma, including $k=0$. □

Two robust Laguerre tests

Let $\rho(dx) = e^{-x} dx$ on $[0, \infty)$. Its normalized moments are $\int x^k/k! d\rho = 1$. Write

$$L_j(x) = \sum_{u=0}^j (-1)^u \binom{j}{u} \frac{x^u}{u!}.$$

These Laguerre polynomials are orthonormal for ρ , and

$$xL_j = (2j+1)L_j - (j+1)L_{j+1} - jL_{j-1}, \quad (5)$$

with the last term absent for $j=0$. Both facts follow from the displayed formula: orthogonality also follows by j integrations by parts in $L_j(x) = e^x(j!)^{-1}(d/dx)^j(e^{-x}x^j)$, and the norm is one by comparison with its leading coefficient.

For a polynomial $P(x) = \sum_s b_s x^s$, put $\|P\|_* := \sum_s |b_s| s!$. If a positive measure ν has normalized-moment errors at most δ through degree N , then

$$\left| \int P d\nu - \int P d\rho \right| \leq \delta \|P\|_* \quad (\deg P \leq N). \quad (6)$$

Furthermore,

$$\|L_i L_j\|_* \leq 3^{i+j}, \quad \|xP\|_* \leq (\deg P + 1) \|P\|_*. \quad (7)$$

For the first bound, expand the two polynomials and use $(u+v)!/(u!v!) \leq 2^{u+v}$.

Lemma 4. *Suppose $d \geq 2$ and ν is a finite positive measure on $[0, \infty)$ whose normalized moments through degree $2d-1$ differ from those of ρ by at most δ . If*

$$\delta d 9^d < \frac{1}{6},$$

then ν has positive mass in $[0, 2/d]$. Moreover, every atom of ν at a point $x \in [0, 2/d]$ has mass at most

$$\frac{32}{d} + \frac{1024\delta 9^d}{d^2}.$$

Proof. Use the tapered polynomial

$$q_d(x) = \sum_{j=0}^{d-1} \left(1 - \frac{j}{d}\right) L_j(x).$$

Orthonormality and (5) give

$$\begin{aligned} \int q_d^2 d\rho &= \frac{(d+1)(2d+1)}{6d}, \\ \int x q_d^2 d\rho &= \sum_{j=1}^d j \left(\frac{1}{d}\right)^2 = \frac{d+1}{2d}. \end{aligned}$$

Thus for $R_d(x) = (2/d - x)q_d(x)^2$,

$$\int R_d d\rho = \frac{(d+1)(d+2)}{6d^2} > \frac{1}{6}.$$

By (7), $\|q_d^2\|_* \leq (\sum_{j < d} 3^j)^2 < 9^d/4$, and hence $\|R_d\|_* \leq d 9^d$ for $d \geq 2$. Equation (6) therefore gives $\int R_d d\nu > 0$. Since $R_d \leq 0$ on $[2/d, \infty)$, some positive mass lies below $2/d$.

For the atom bound, fix $x \in [0, 2/d]$ and set $r = \lfloor d/8 \rfloor$. For $0 \leq j \leq r$, the coefficient formula yields

$$|L_j(x) - 1| \leq \sum_{u \geq 1} \frac{(jx)^u}{u!} \leq e^{1/4} - 1 < \frac{1}{2}.$$

Also $|L_j(x)| \leq e^{1/4} < 2$. Therefore

$$S_x := \sum_{j=0}^r L_j(x)^2 \geq \frac{r+1}{4} \geq \frac{d}{32}.$$

The polynomial

$$P_x(y) = \frac{1}{S_x} \sum_{j=0}^r L_j(x) L_j(y)$$

satisfies $P_x(x) = 1$ and $\int P_x^2 d\rho = 1/S_x \leq 32/d$. Its coefficient norm satisfies

$$\|P_x^2\|_* \leq \frac{1}{S_x^2} \left(\sum_{j=0}^r |L_j(x)| 3^j \right)^2 \leq \frac{1024}{d^2} 9^{r+1} \leq \frac{1024}{d^2} 9^d.$$

The mass of an atom at x is at most $\int P_x^2 d\nu$. Apply (6) to finish the proof. $\square$

Proof of Theorem 1. Take

$$d = \left\lfloor \frac{\log m}{8 \log 3} \right\rfloor, \quad \nu = \sum_{q \in G} \eta(q) \delta_{\lambda(q)}.$$

For all sufficiently large m , $d \geq 2$ and $2d - 1 \leq \sqrt{m}$. Lemma 3 gives the normalized-moment error $\delta \leq 4d/\sqrt{m}$ through degree $2d - 1$. Since $9^d \leq m^{1/4}$,

$$\delta d 9^d \leq 4d^2 m^{-1/4} \longrightarrow 0, \quad \frac{1024 \delta 9^d}{d^2} \leq \frac{4096 m^{-1/4}}{d} = o(1/d).$$

Lemma 4 supplies a good row with $\lambda < 2/d$. All odds decrease with the row, so the last row is good and has $\lambda(K) < 2/d$. Its η -mass is bounded by the mass of the corresponding atom of ν , and therefore is at most $33/d$ for all sufficiently large m . Finally,

$$A(K) = \prod_j \frac{1}{1 + r_j(K)} \geq e^{-\lambda(K)} \geq e^{-2/d}.$$

Consequently

$$(m+1)w_K e^{-2/d} \leq \eta(K) \leq \frac{33}{d}.$$

Since $d \asymp \log m$, this proves the last-row bound. Reversing row order and replacing every p_j by $1 - p_j$ preserves (1) (by expansion, or by uniform comparison patterns), so it also proves the first-row bound. Equal weights now give $K \geq cm \log m$. $\square$

Scope. The argument uses only the monotone local comparison tensor. It needs neither rationality of its entries nor arithmetic divisibility. The construction with $O(n^2)$ faces on one die shows that a substantially stronger lower bound, if true, must remain compatible with that scale.